

Basic Tissues

TISSUE [Fr. Tissu, Woven; L. Texo, to Weave]

- is a group of structurally and functionally closely related cells and their product (intercellular substance = ECM) specialized to perform specific function(s).
- Tissues are made of two interacting components: **cells** and intercellular substance known as **extracellular matrix (ECM)**.
- The **ECM** consists of many kinds of molecules, including **fibers**, **ground substance** (glycosaminoglycans, proteoglycans, glycoproteins) & **tissue fluid**.
- **Fibrillar ECM components usually organize and** form complex structures, such as collagenous and elastic fibers bundles, external laminae and basement membranes.

The Four fundamental types of tissues are: **epithelial tissue, connective tissue, muscle tissue, and nervous tissue.**

– **Each of the fundamental tissues:**

- is formed by several types of related cells and specific associations of cells and the extracellular matrix
- exists not as isolated units but rather in association with one another and in variable proportions, forming different organs and systems of the body.

– These characteristic associations facilitate the recognition of the many subtypes of tissues.

The four basic types of tissues from superficial to deep are:

- 1.Epithelial tissue**
- 2.Connective tissue**
- 3.Nervous tissue**
- 4.Muscle tissue**

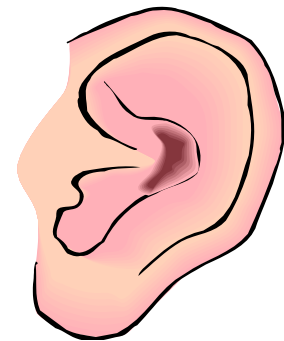
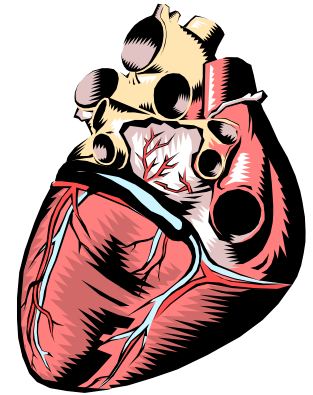
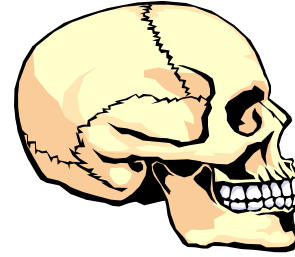
Objectives

- **Characterizing and identifying various types of tissues of the human body**
 1. **Epithelial tissue**
 2. **Connective tissue**
 3. **Muscular tissue**
 4. **Nervous tissue**

Classification of Human Tissues

- Epithelial tissue
- Connective tissue
- Muscular tissue
- Nervous tissue

Tissues



- There are four primary tissue types in the human body:
 1. Epithelial (*membranous and glandular*)
 2. Connective (*supportive tissue*)
 3. Muscle (*movement*)
 4. Nervous (*control/communication*)
- These tissues differ in the types and functions of their cells, and the products of those cells and the relative distribution of the two.
- Which picture on this slide corresponds to which tissue?

4 TYPES of TISSUE

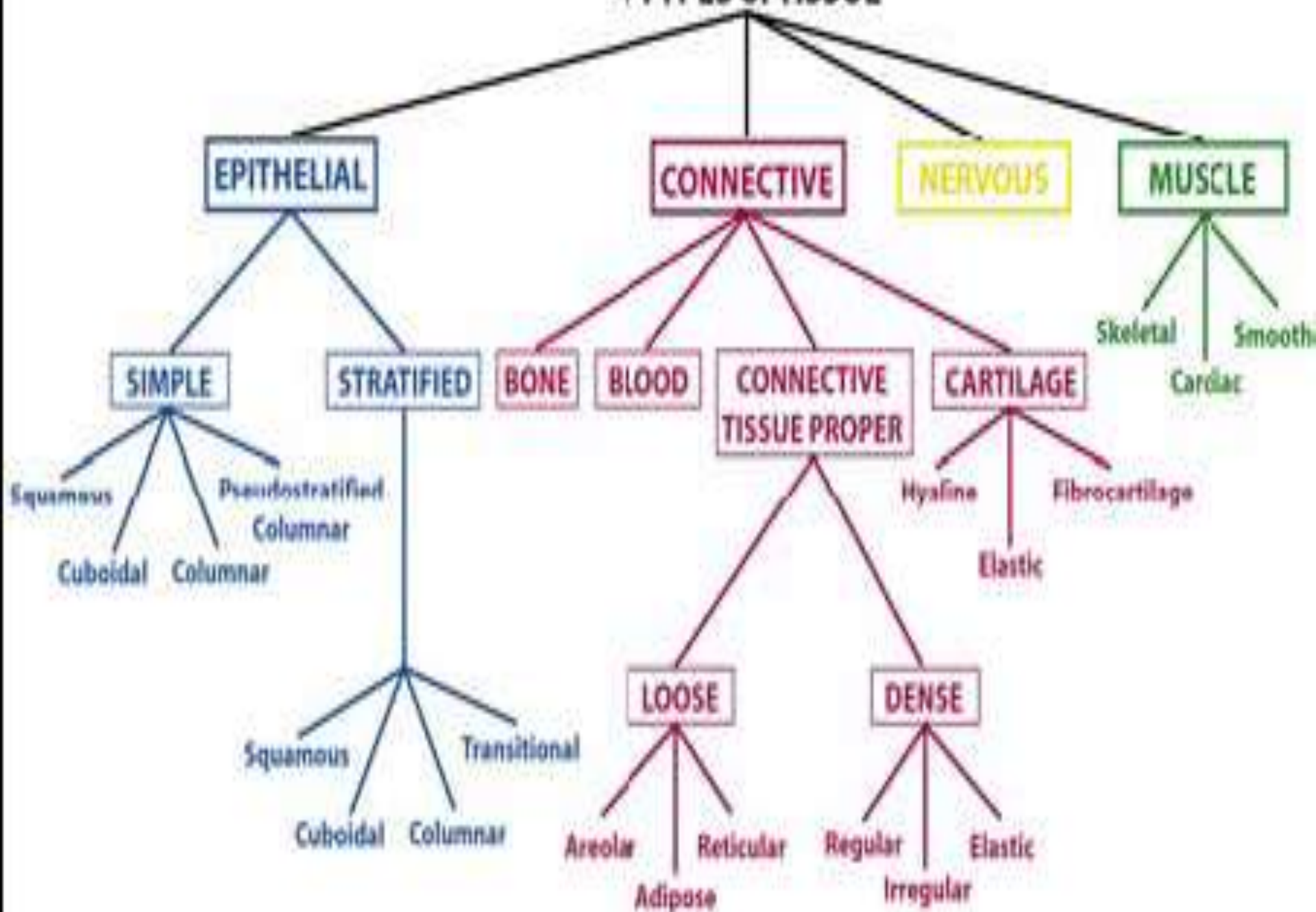
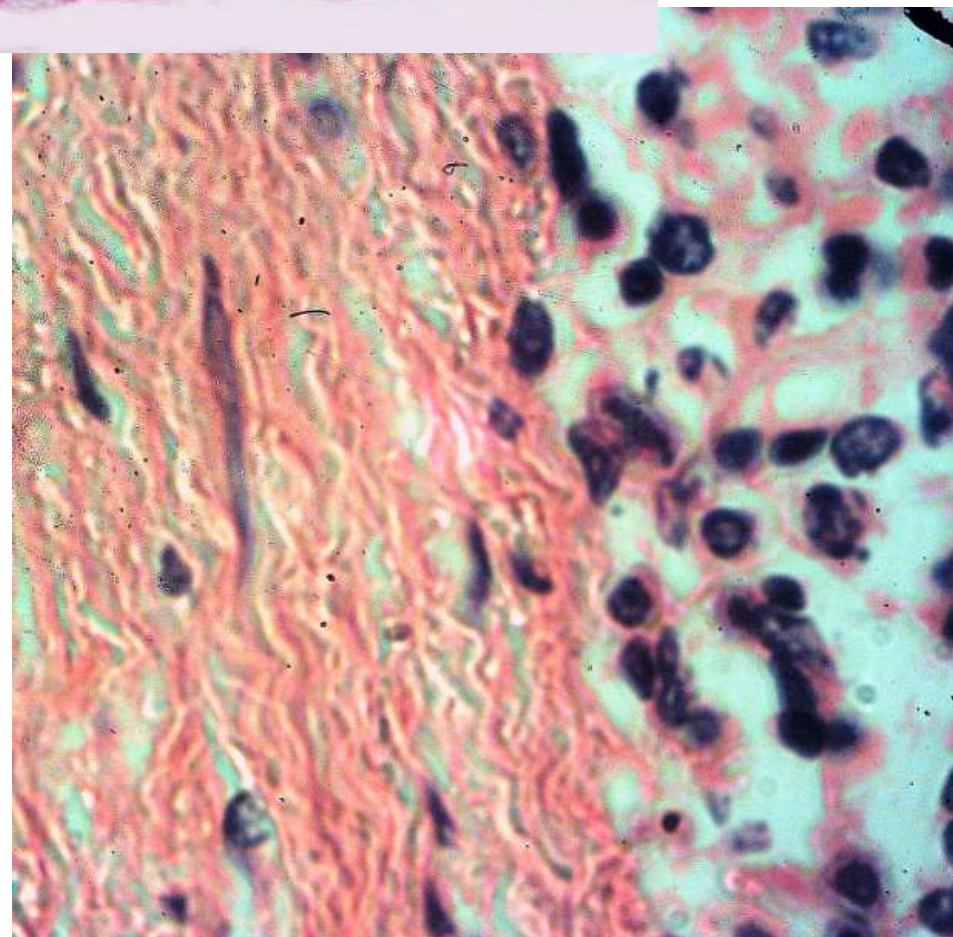
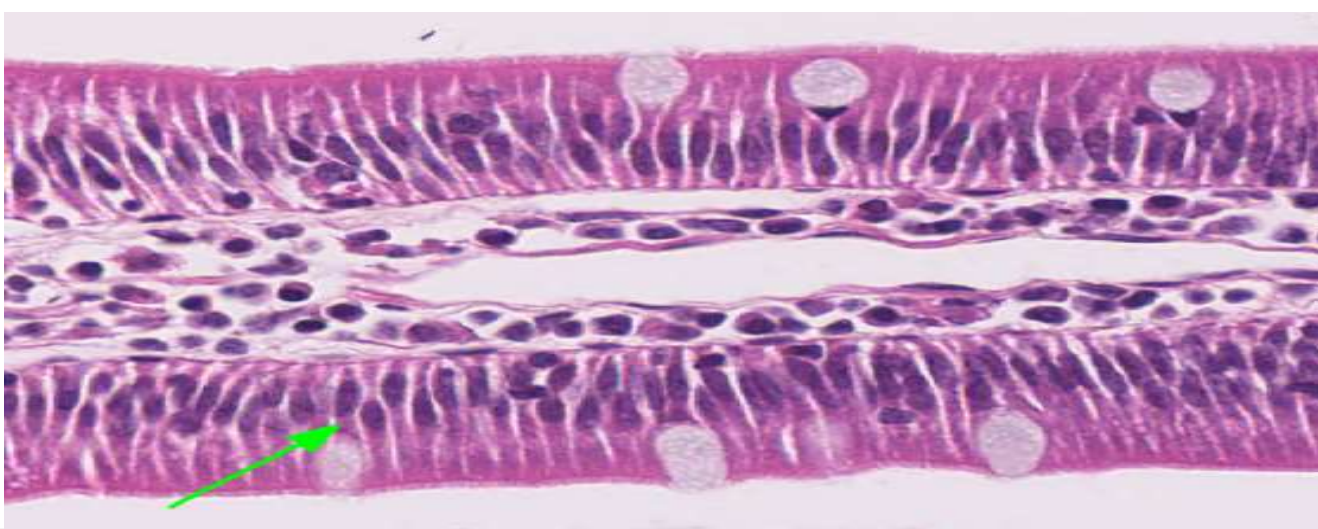
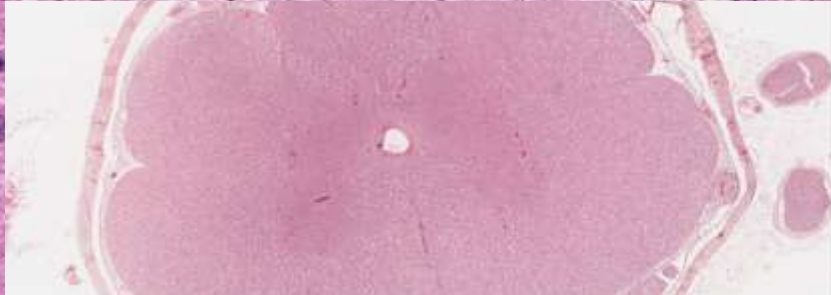
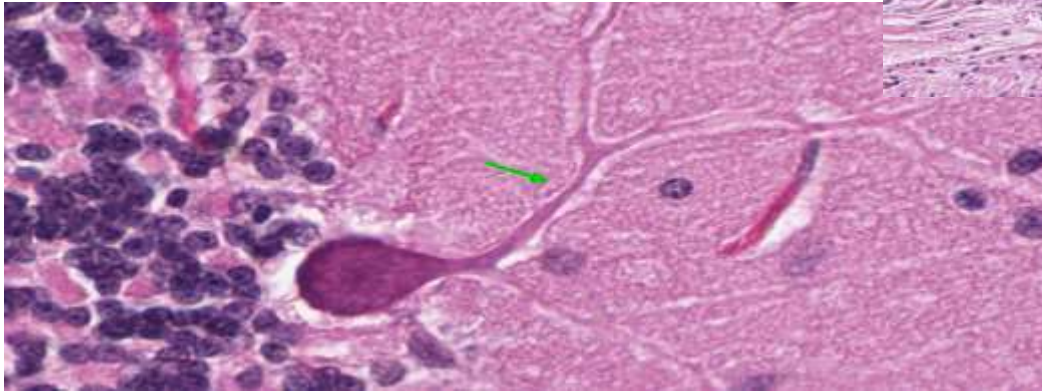
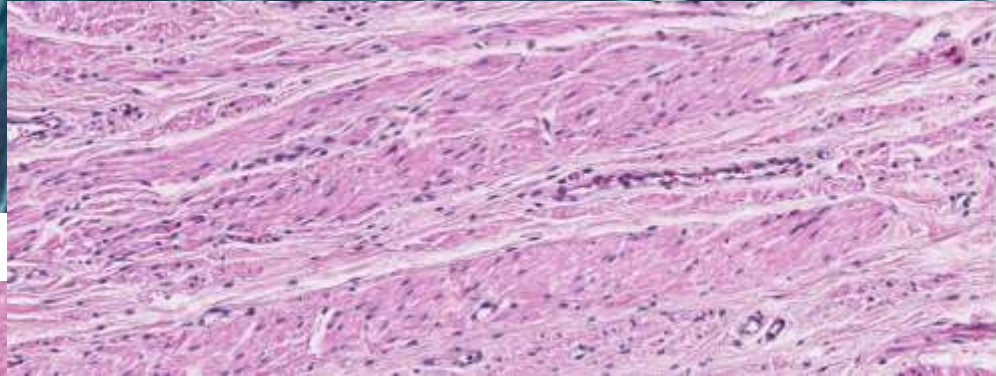
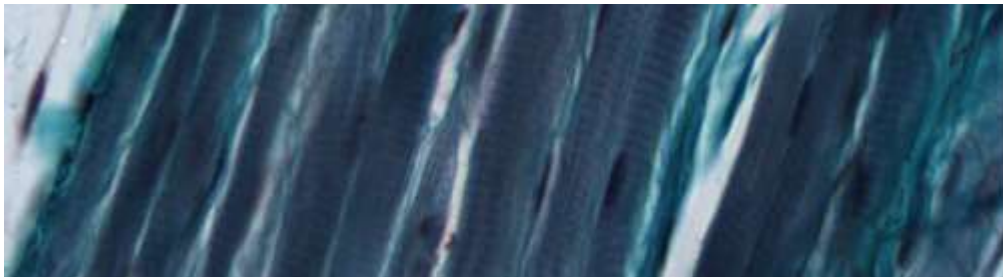
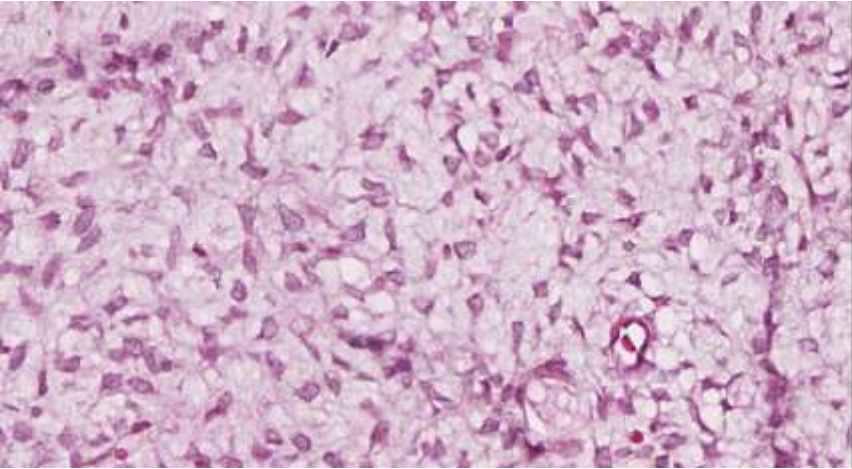
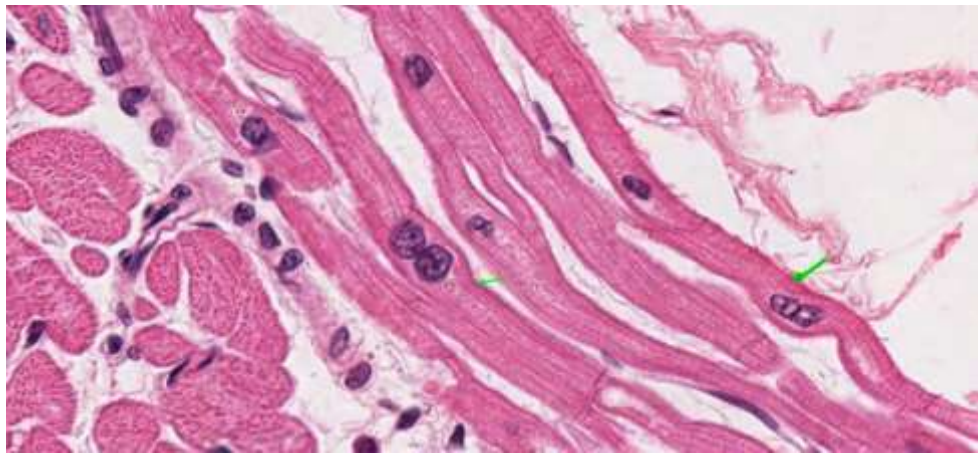
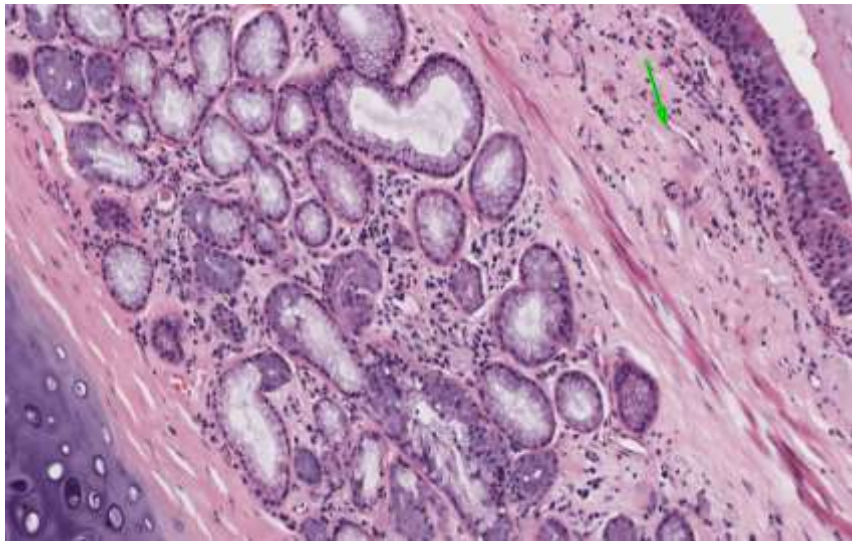


Table. Main Characteristics of the Four Basic Types of Tissues

Basic Tissues	Cells	Extracellular Matrix	Main Functions
Epithelial	Closely arranged polarized cells	Very small amount	Covering body surfaces, Lining tubes or body cavities, glandular secretion
Connective	Several types of fixed and wandering cells	Abundant amount	Support and protection
Muscle	Elongated contractile cells	Moderate amount	Movement
Nervous	Cells with Intertwining elongated processes	Lacks fibers, has only gelatinous ground substance	Transmission of nervous impulses





1. EPITHELIAL TISSUE/EPITHELIUM, εpi'θi:liam

Introduction:

- (Gk. *epi*, upon, + *thele*, "**teat, nipple**")
- is an avascular tissue composed of contagious cells that cover **body surfaces** or line **closed cavities and tubes** that communicate with the exterior (the alimentary, respiratory, and genitourinary tracts).
- formed by a group of closely arranged cells with very little intercellular substance (ECM).
- consists of continuous sheet of cells forming membranes which cover or line body structures or grouped as three-dimensional secretory units of glands.
- the cells are highly cohesive,

Epithelial tissue

Epithelial tissue is composed of crowded cells with very little intercellular substance. Cells are attached together by cell junctions

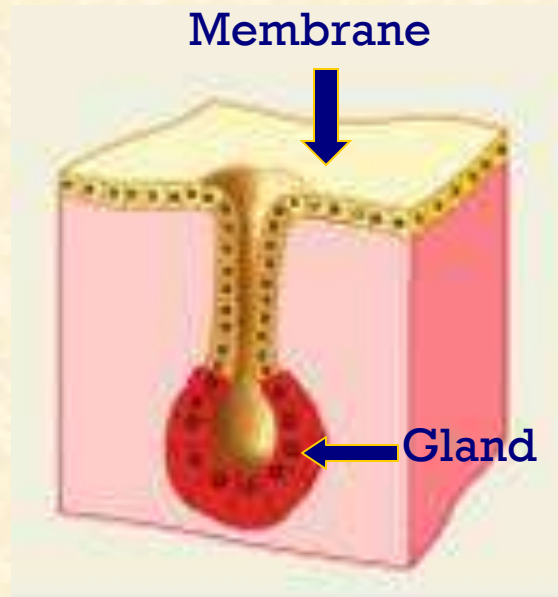
Epithelial tissue forms **membranes** that cover body surfaces or line body cavities

Epithelial tissue form **glands** [Exocrine OR Endocrine glands]

Epithelial tissue is **avascular** (NOT penetrated by blood and lymphatic vessels)

Function of Epithelium:

- a) Secretion
- b) Protection
- c) Absorption
- d) Gas exchange



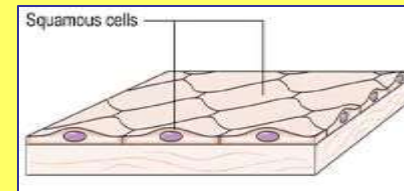
Types of epithelial Tissues:

Simple Epithelia (Formed of one row of cells)

According to the shape of cells

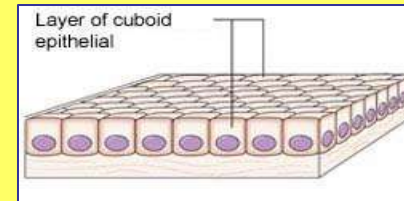
Simple squamous epithelium

Single row of flattened cells with flattened nuclei rest on basement membrane



Simple cubical epithelium

Single row of cubical cells with rounded nuclei rest on basement membrane

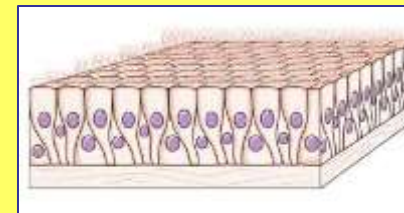


Simple columnar epithelium

Single row of columnar cells with oval basal nuclei rest on basement membrane



Pseudostratified columnar epithelium



Stratified epithelia

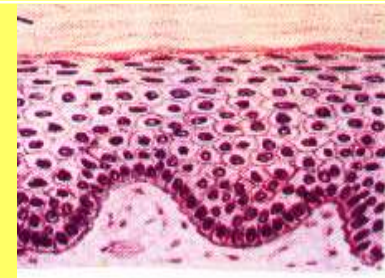
(Formed of two or more rows of cells)

According to the shape of surface cells

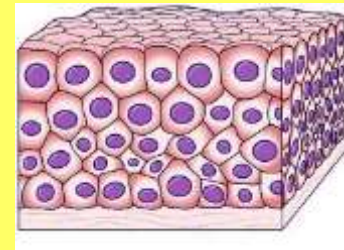
Stratified squamous epithelium



Stratified squamous keratinized epithelium

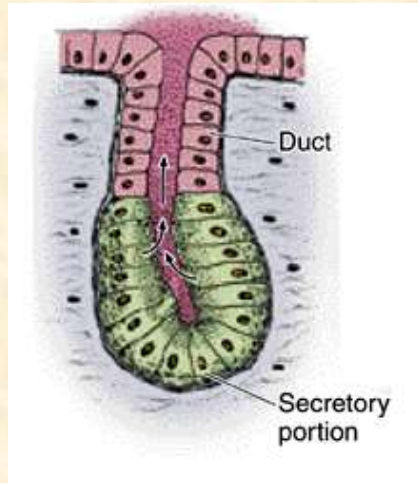


Transitional epithelium



Glandular epithelium:

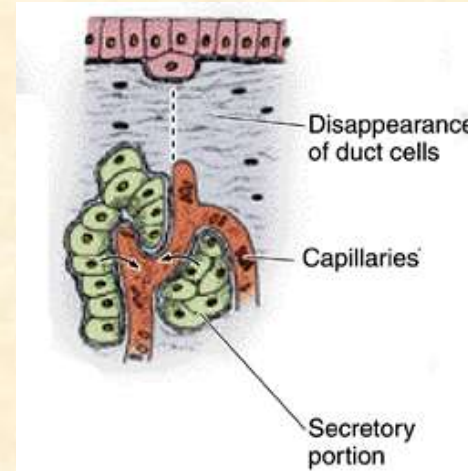
Exocrine gland



Release secretion into the surface through **a Duct**

Example: Salivary glands

Endocrine gland



Have **NO Duct**

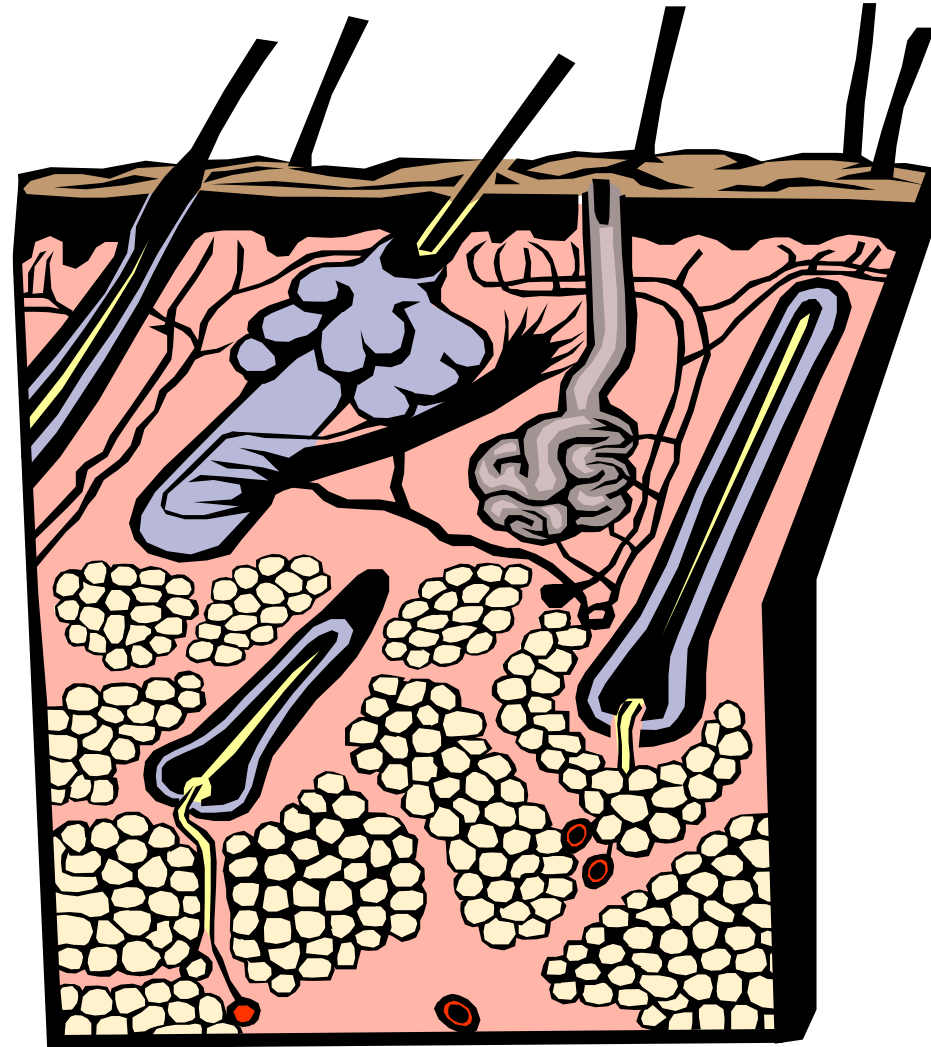
Release hormones

Released into blood stream

Example: Pituitary gland, thyroid gland,.....

Functions of Epithelial Tissue

- Covers the surface of the body, lines the body cavities, forms the external and internal linings of most organs, and constitutes the bulk of most glands.
- It's found all over the body!
- **Functions include:**
 - 1. Protection**
 - 2. Absorption**
 - 3. Secretion**
 - 4. Filtration and diffusion**
 - 5. Sensory reception**
 - 6. Transcellular and surface transport**



- Above we have a section of skin. Where would the bulk of the epithelial tissue be found?

The principal functions of epithelial tissues are

- 1. Covering and lining of surfaces (e.g., skin covers the entire body, epithelial lining of intestines),**
 - **Provides Protection**--- from UV rays (e.g. Melanin), loss of fluid or desiccation (corneum), infection, friction (mechanical abrasion), etc.
- 2. Transcellular transport (transcytosis):**
 - Diffusion of gases (oxygen & carbon dioxide) across the epithelial cells of lung alveoli & capillaries.
 - Carrier protein-mediated transport (e.g. digested food such as amino acids and glucose across intestinal epithelia)
 - Vesicle-mediated transport (of IgA & other molecules)
- 3. Surface transport by cilia** (e.g. mucus in respiratory system, ovum in uterus and oviduct)

Functions of epithelial tissues (Cont'ed)

4. **Absorption**: occurs via endocytosis or pinocytosis in various organs (e.g., the proximal convoluted tubules of kidneys)
5. **Secretion**: of various products (e.g., hormones, mucus, sweat, saliva, proteins, etc) by exocytosis,
6. **Sensory functions** (e.g., gustatory and olfactory neuroepithelium)
7. **Contractility**: e.g., **myoepithelial cells**
 - epithelial cells specialized for high contractility
 - found in sweat, salivary, lacrimal, & mammary glands to assist the process of their secretion mechanically.

Apical Modifications of Epithelial cells for different functions

–The free or apical surface of many types of epithelial cells has specialized structures to increase the cell surface area or to move substances or particles on the surface.

1. Cilia

–are elongated, highly motile cytoplasmic extensions of almost all cells, 2–10 μm long and 0.05–0.25 μm (50–250 nm) in diameter, which are much longer & two times wider than a typical microvillus.

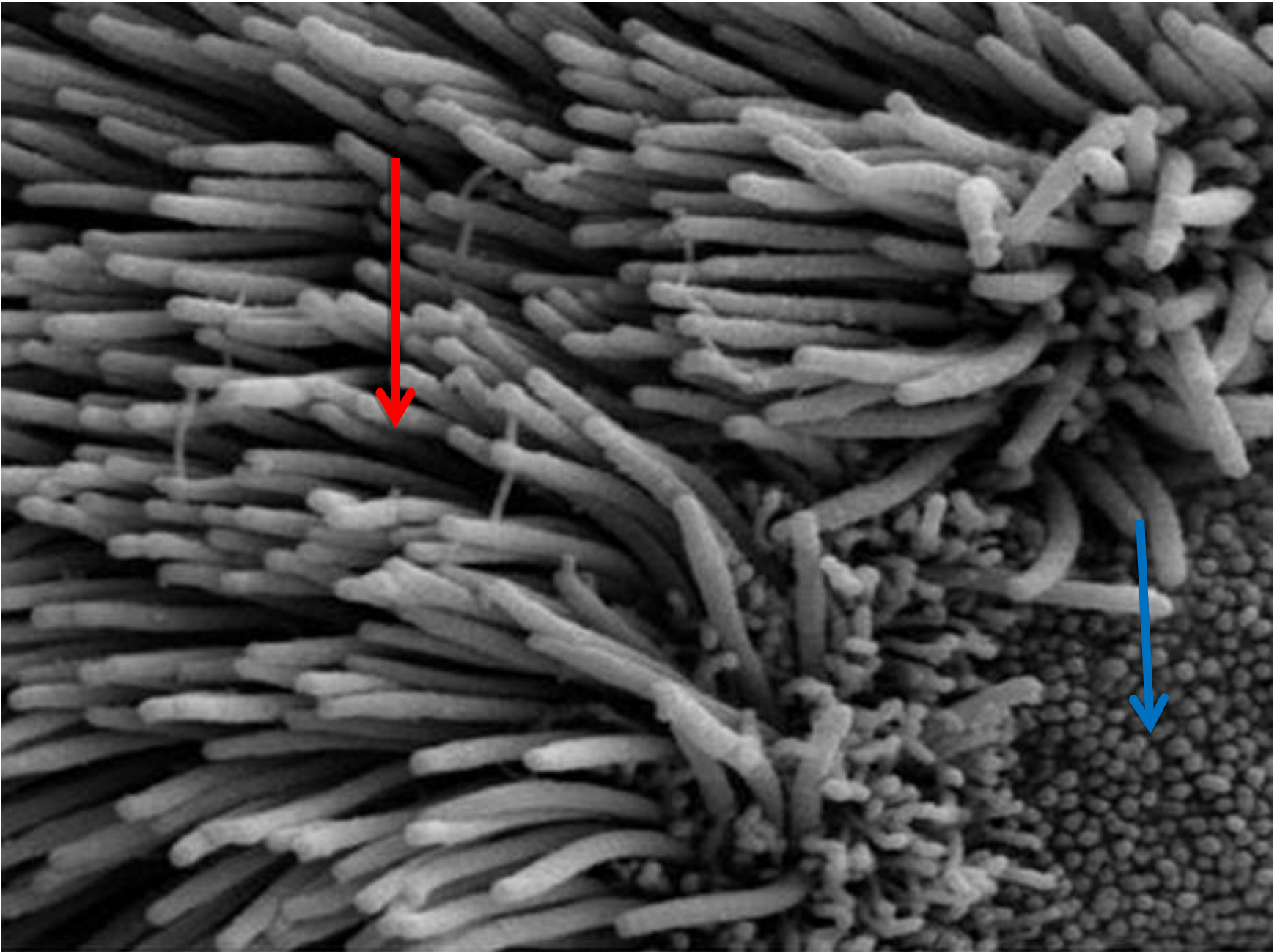
–each cilium is bounded by the cell membrane and contains an axoneme with a central pair of microtubules surrounded by nine peripheral microtubular pairs.

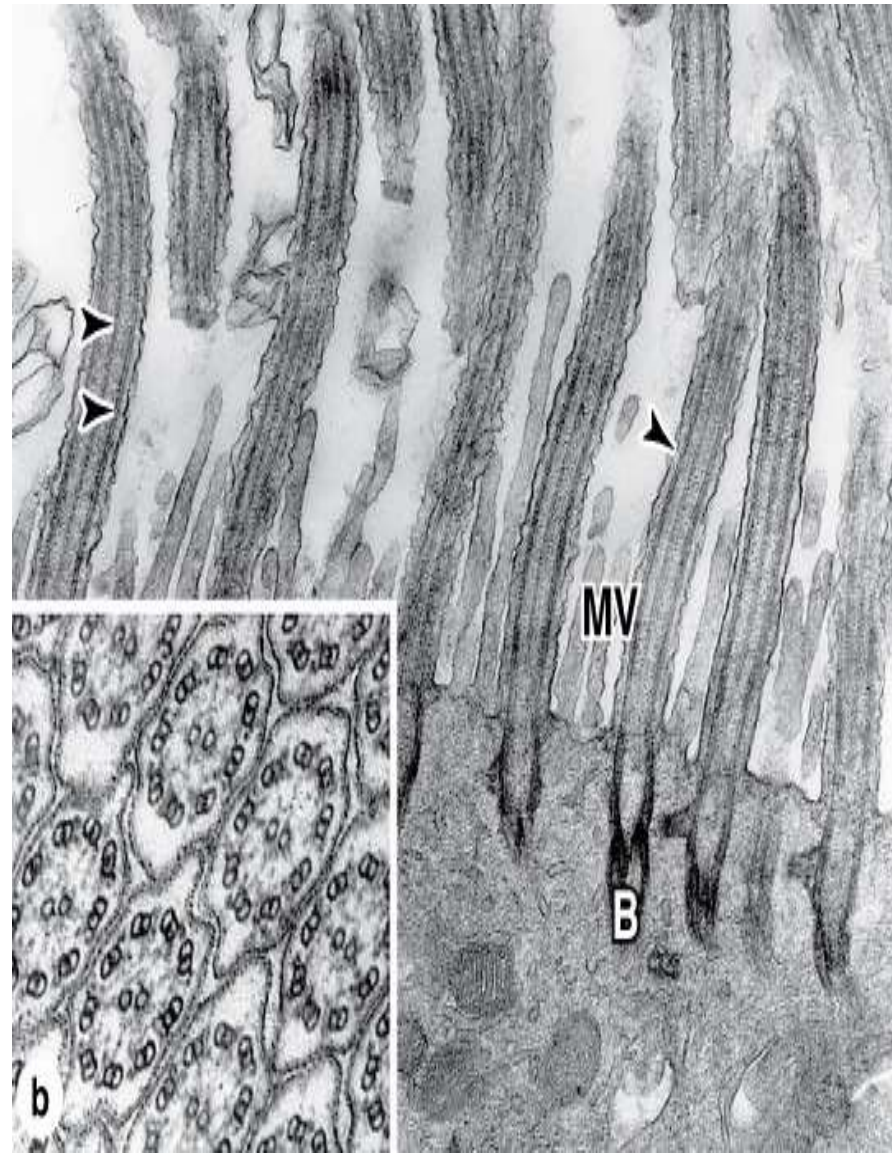
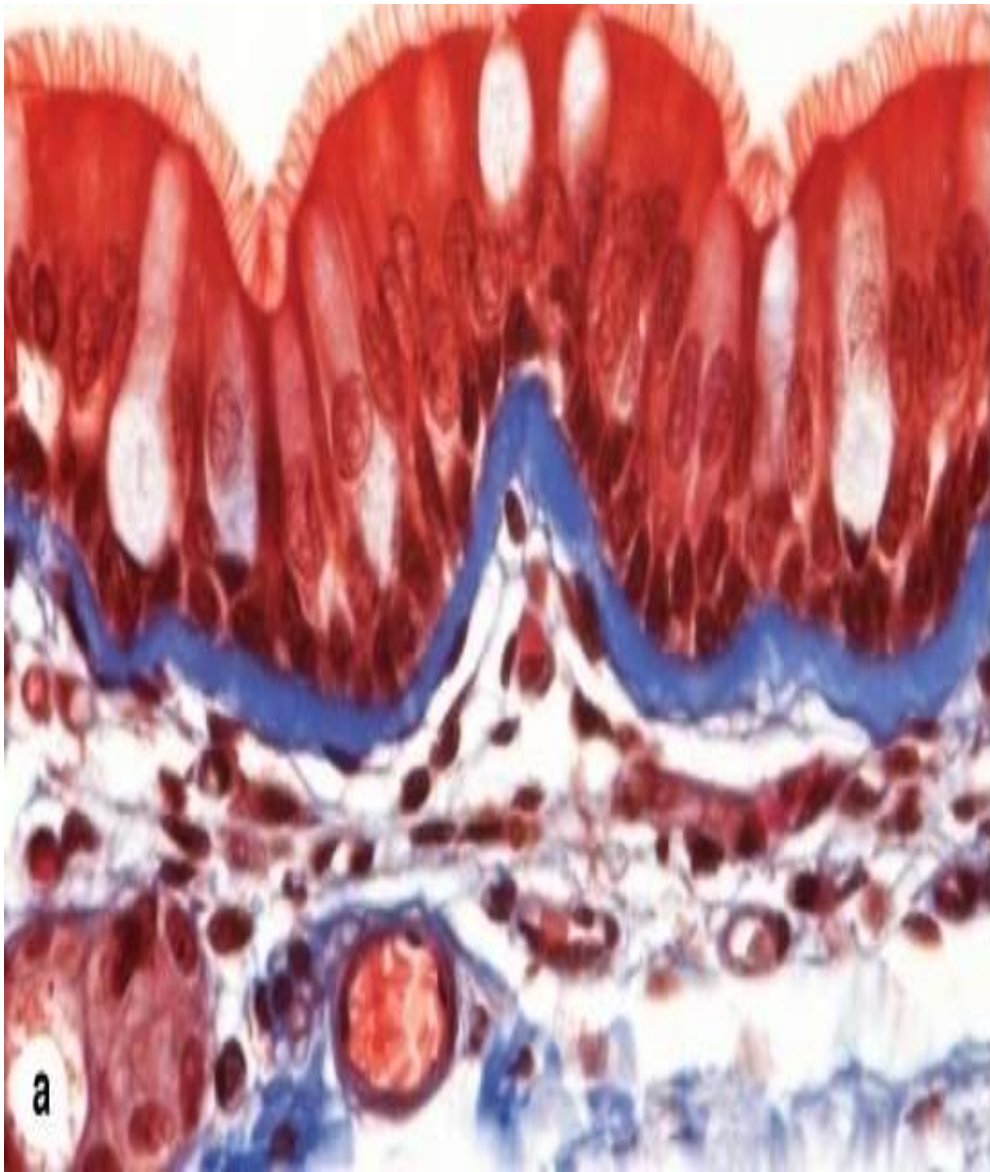
–Originate from the **basal bodies**, which are electron-dense structures in the apical cytoplasm at the apical pole just at the roots of the cilia.

Cilia

- Basal bodies have a structure similar to that of centrioles.
- In living organisms, cilia exhibit rapid back-and-forth movements coordinated to propel a **current of fluid and suspended particles** in one direction over the ciliated epithelium.
- The motion occurs due to activity of **ciliary dynein** present on the peripheral microtubular doublets of the axoneme, with **adenosine triphosphate (ATP)** as the energy source.
- A ciliated lining cell of the trachea can bear 250 to 300 cilia.
- Three types of cilia: **Typical cilia** (perfectly motile), **primary** (nonmotile) and **nodal cilia** (motile).

Cilia (red arrow) and Microvilli (blue arrow) on a tracheal cell



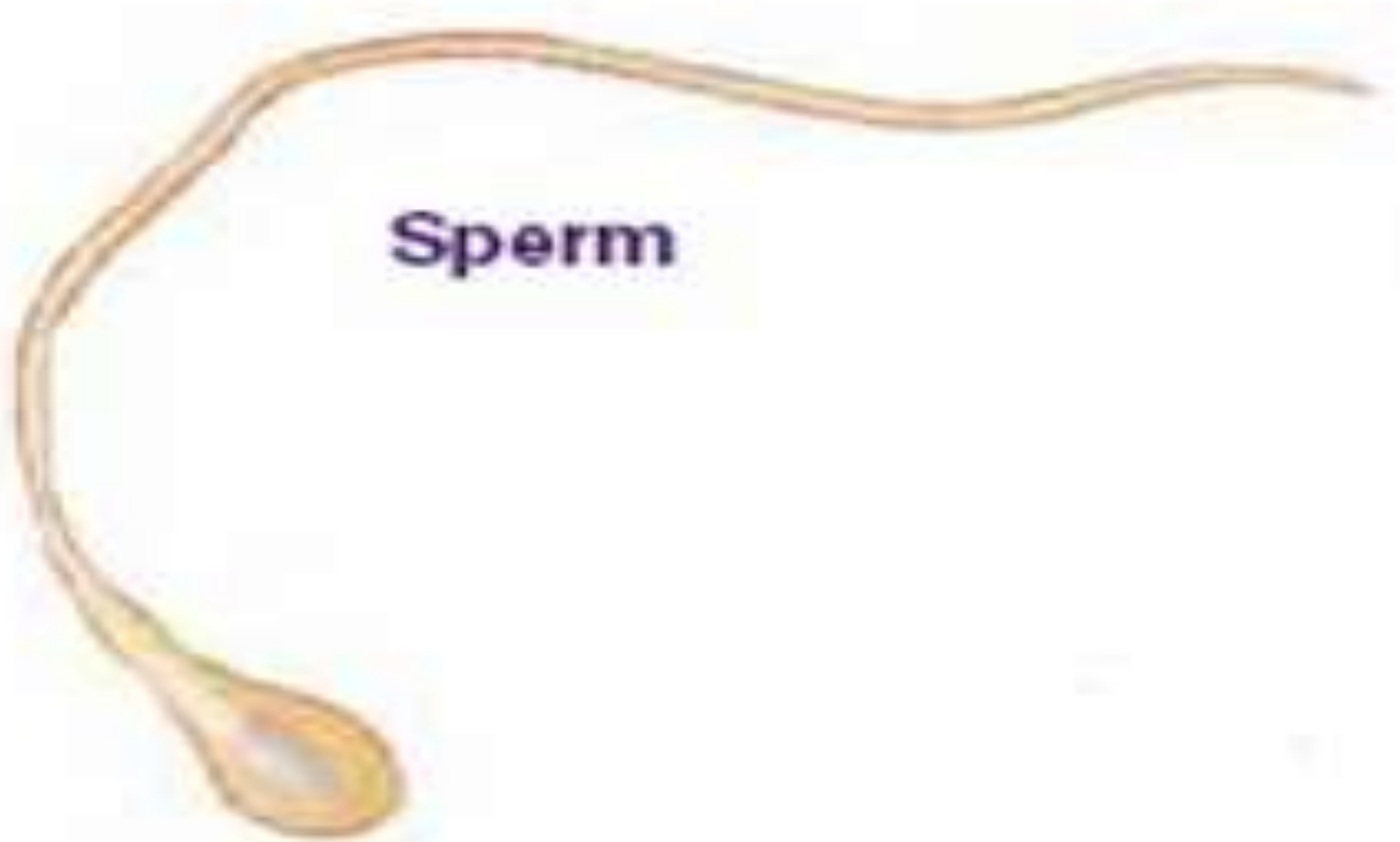


Ciliated (arrow head) lining epithelium of the trachea and 9+2 axoneme pattern of each cilium

2. Flagella

- present only in spermatozoa
- are apical elongation and modifications of immature cells known as spermatids by the cytodifferentiation known as spermiogenesis.
- are similar in structure to cilia but are much longer and are normally limited to **one flagellum per cell**.
- Sperm cells are the only flagellated human body cells

A human sperm cell



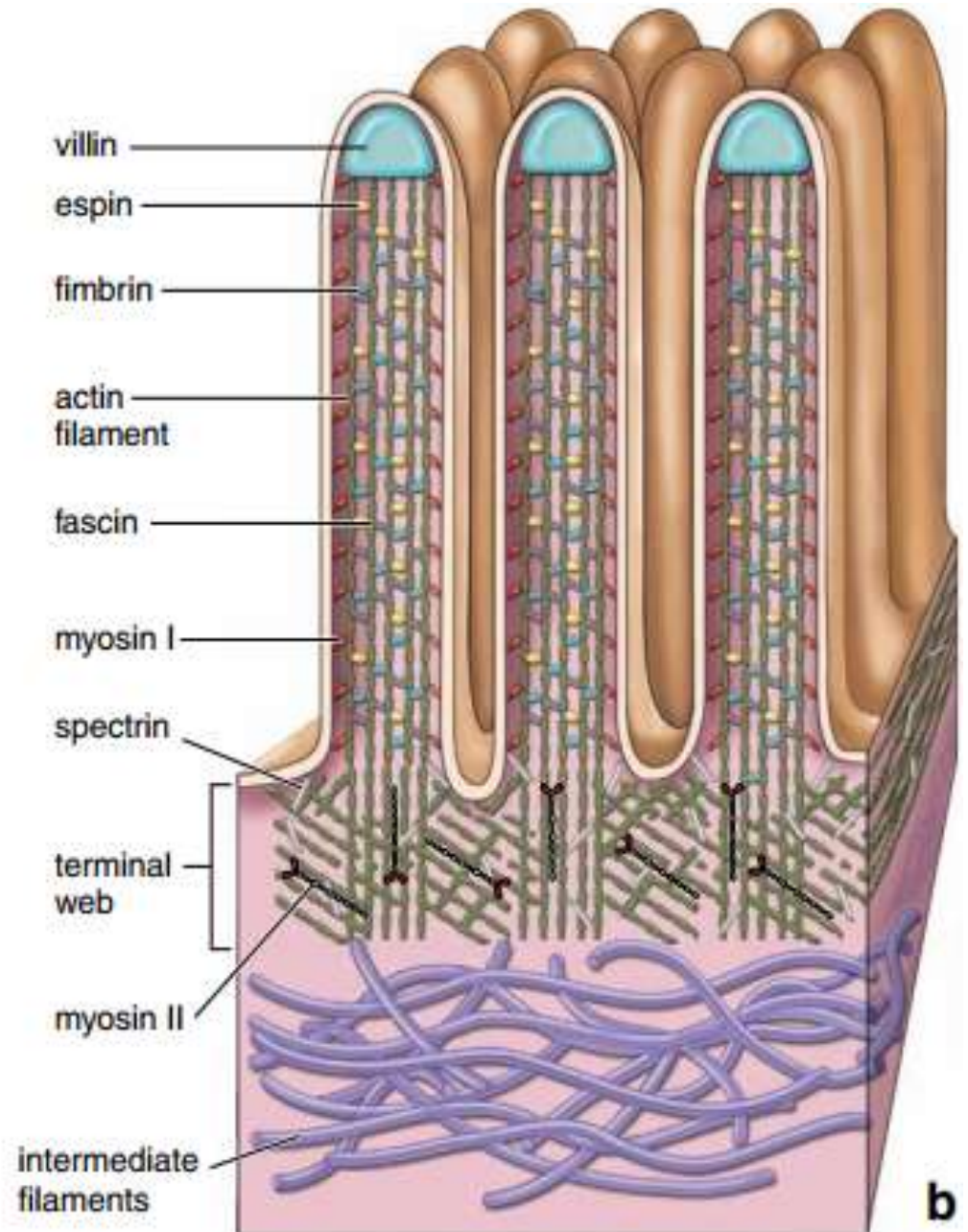
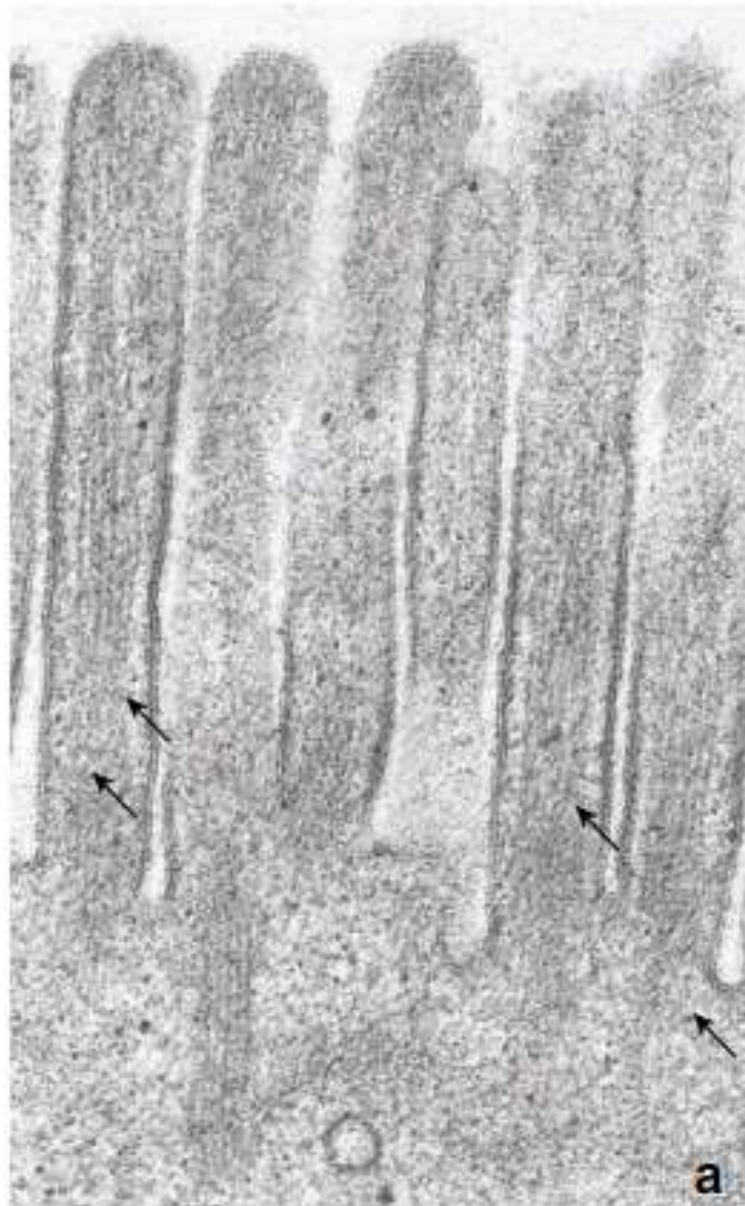
3. Microvilli

- are short or long fingerlike cytoplasmic extensions or folds that pursue a sinuous course, and they range in number from a few to many.
- most are temporary, reflecting cytoplasmic movements and the activity of actin filaments.
- In absorptive cells, such as the lining epithelium of the small intestine, the apical surface presents orderly arrays of many hundreds of more permanent **microvilli** (L. *villus*, tuft), that appear as **striated border**.
- The average microvillus is only about 1-3 μm high and 50-100 nm (0.1 μm) wide, but with hundreds or thousands present on the apical end of each absorptive cell, the total surface area can be increased as much as 20- or 30-fold.

Microvilli

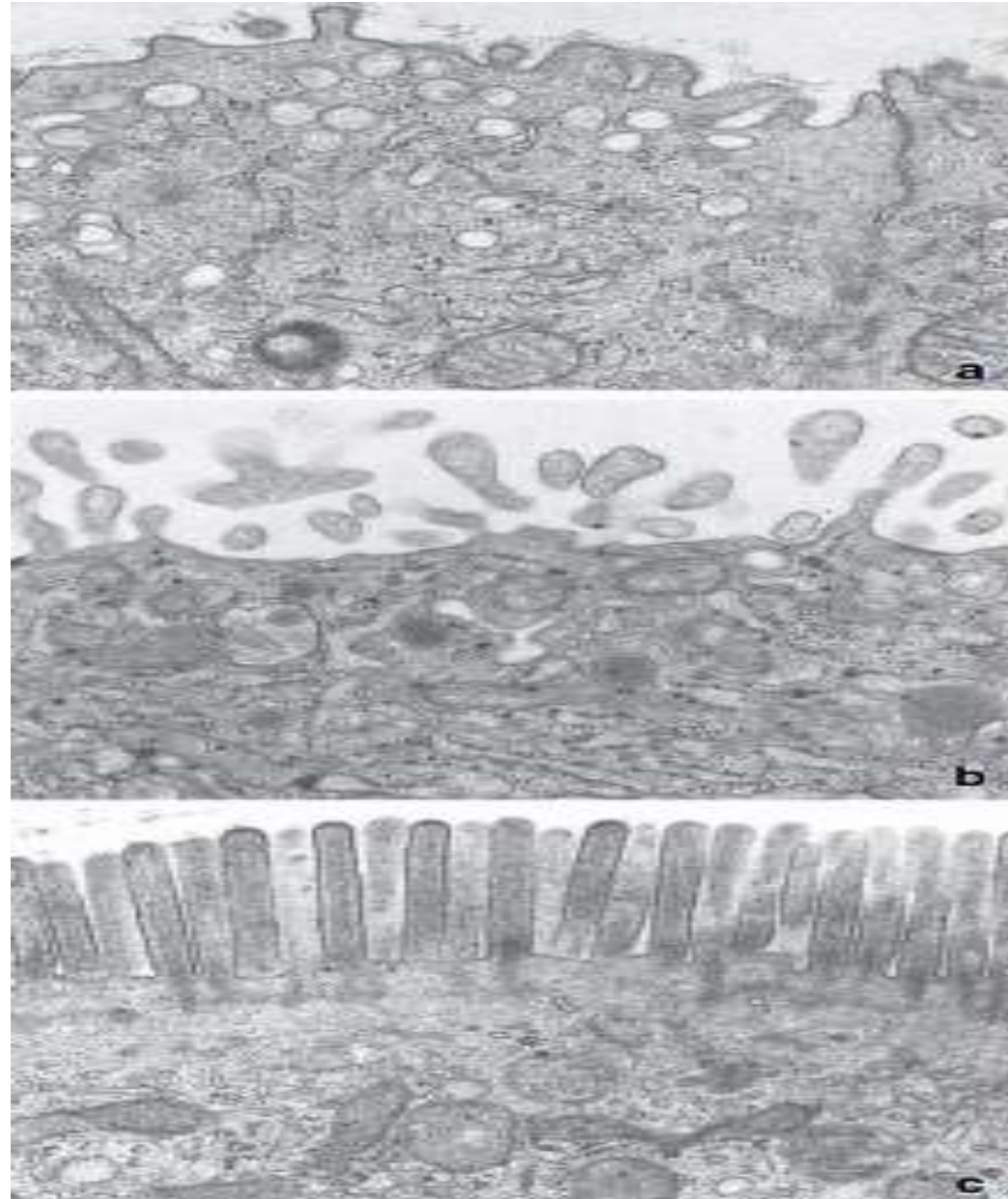
- Each intestinal absorptive cell (enterocyte) can have as many as 3000 microvilli on its apical pole.
- In these absorptive cells the glycocalyx is thicker than that of most cells and includes enzymes for the final stages of certain macromolecules' breakdown.
- The complex of microvilli and glycocalyx is easily seen in the light microscope and is called the **brush** or **striated border**.
- At the root of the microvilli is the **terminal web (cuticular plate)**, a horizontal network of actin microfilaments, tropomyosin and myosin II stabilized by **spectrin** (468 kilodaltons), which also anchors the terminal web to the apical cell membrane.

Molecular structure of microvilli



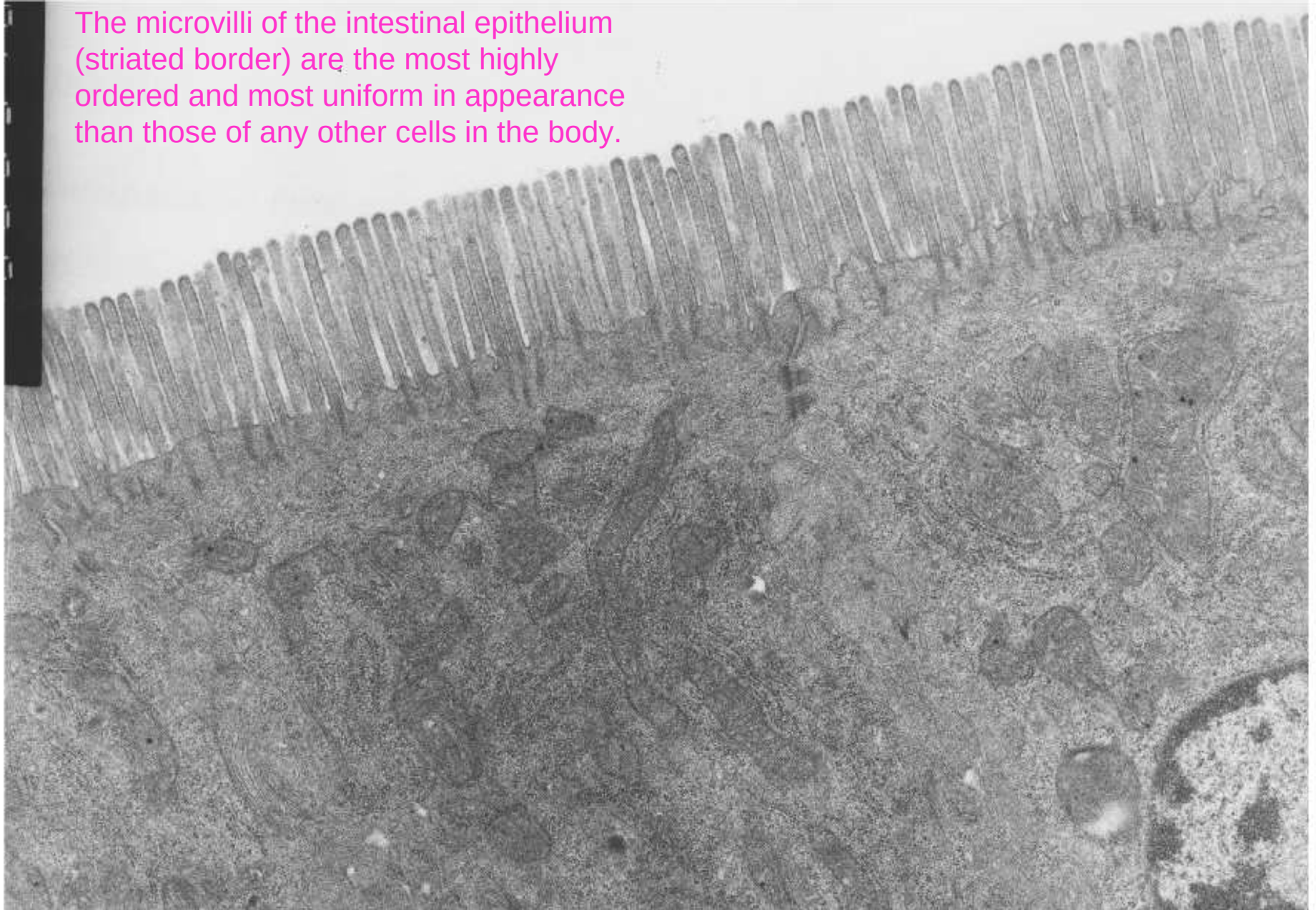
Abundance and size of microvilli on different types of tissues

Electron micrographs showing variation in microvilli of different cell types. a. Epithelial cell of uterine gland; only very few small projections. b. Syncytiotrophoblast of placenta; irregular, branching blunt microvilli. c. Intestinal absorptive cell; uniform, numerous, and regularly arranged microvilli.

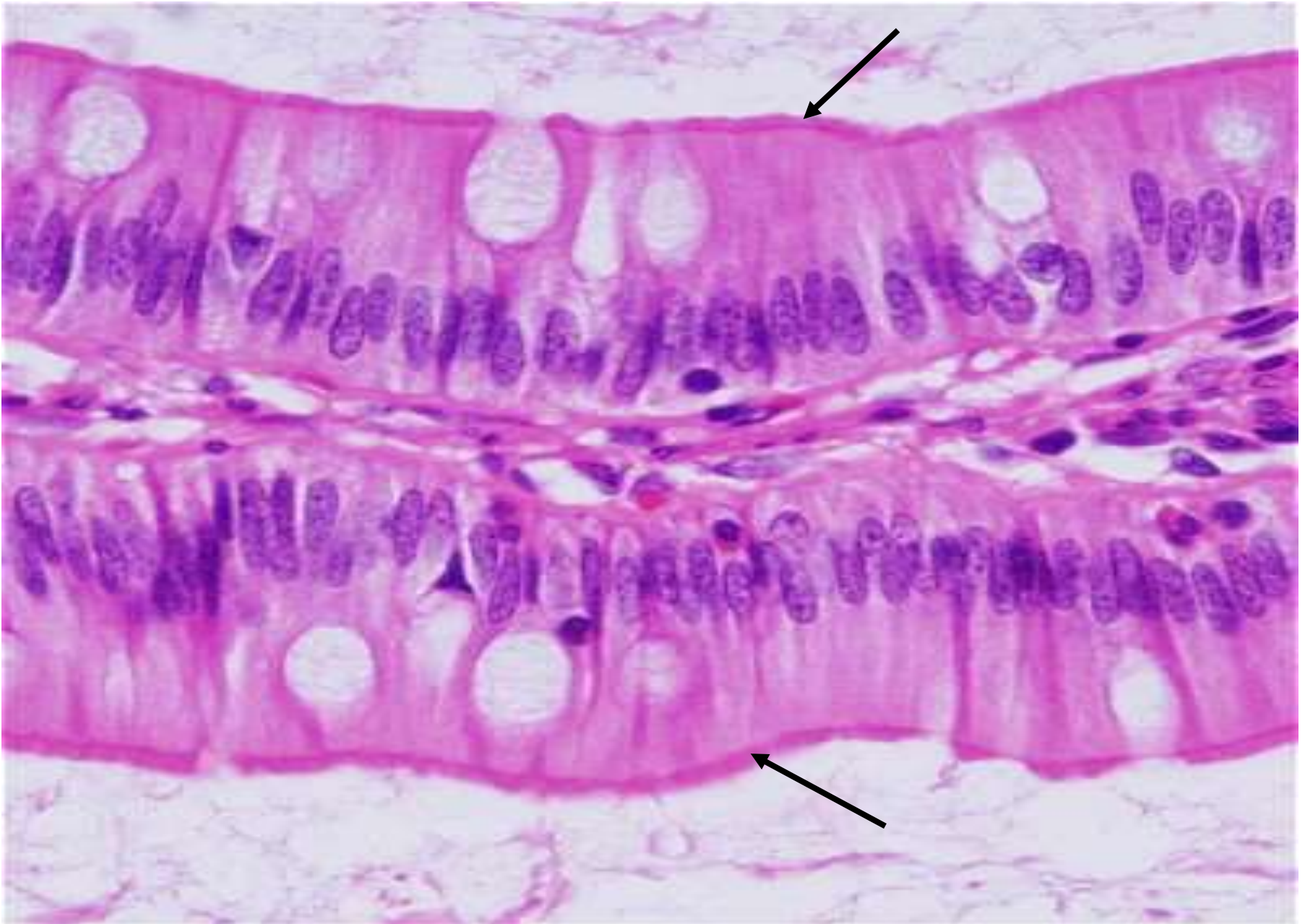


Abundant microvilli at the tip of intestinal absorptive cells

The microvilli of the intestinal epithelium (striated border) are the most highly ordered and most uniform in appearance than those of any other cells in the body.



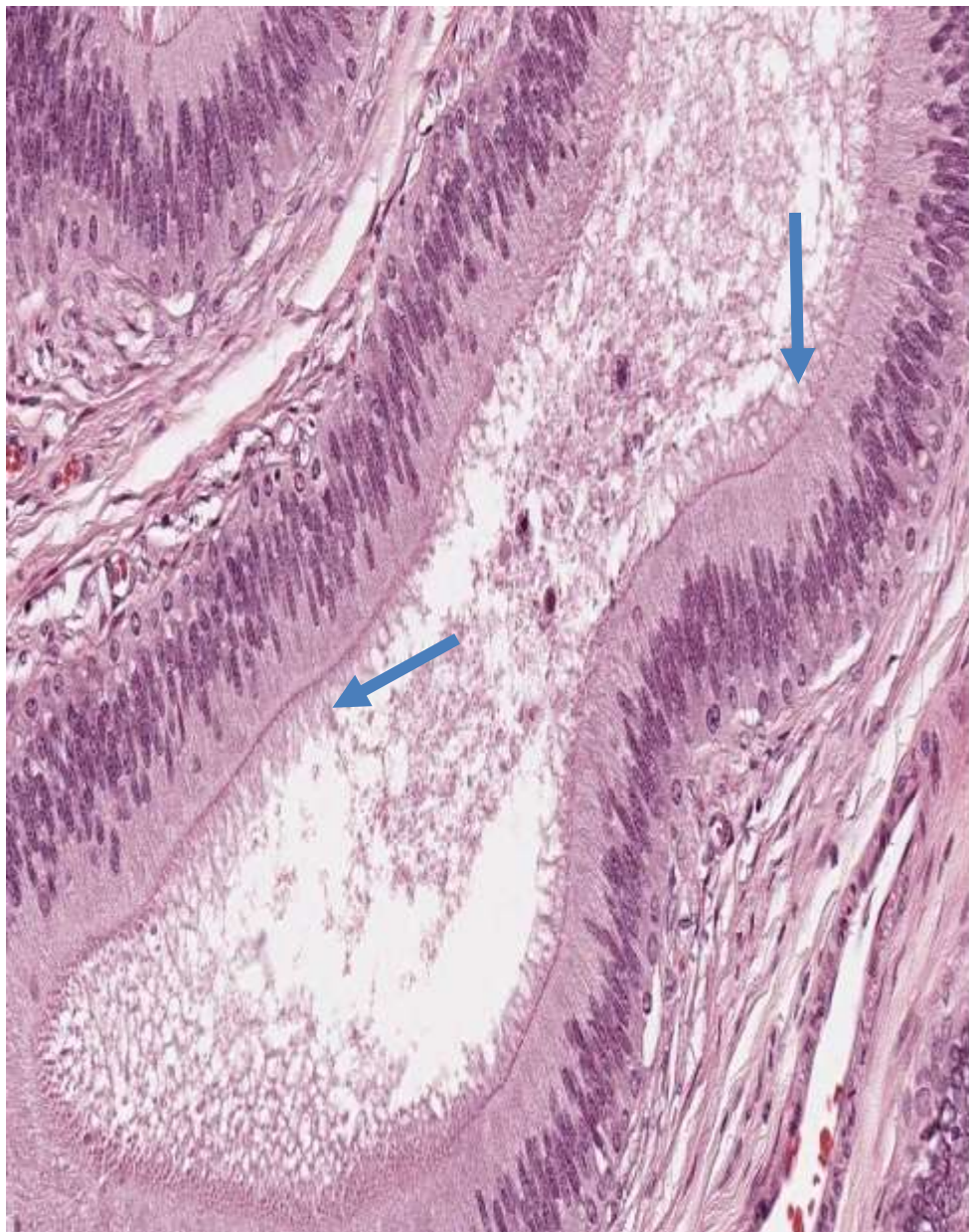
Microvilli appearing brush border on intestinal absorptive cells



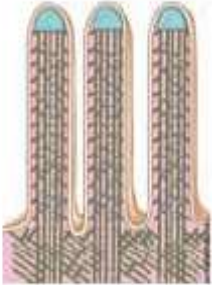
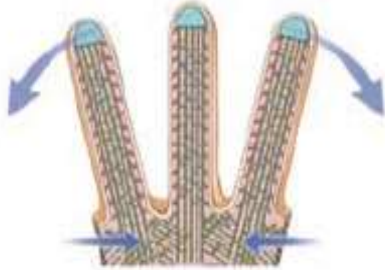
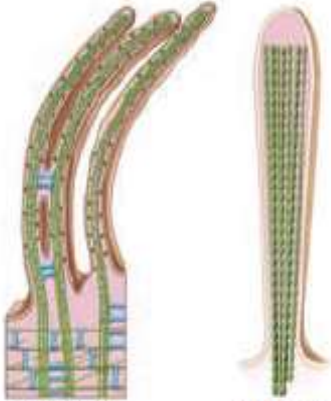
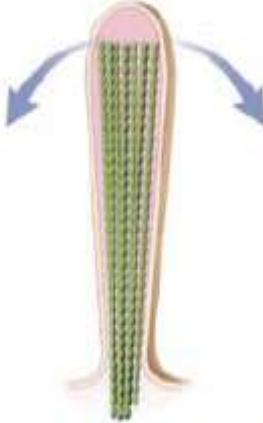
4. Stereocilia/Stereovilli and kinocilia

- are unusually long apical processes of cells (as long as 120 μm) and 100-150 nm in diameter in other absorptive cells such as lining epithelia of epididymis & ductus deferens, and also on hair cells (mechanoreceptors) of inner ear.
- are much longer, sparse and less motile than microvilli, are branched, and should not be confused with true cilia.
- Seen in the light microscope, resemble the hairs of **a paint brush** because of the way they aggregate and **stick together** in tufts of pointed bundles.
- Like microvilli, stereocilia
 - also increase the cells' surface area, facilitating the movement of fluid into and out of the cell.
 - are supported by internal bundles of actin filaments that are cross-linked by fimbrin.

Stereocilia of Male genital ducts



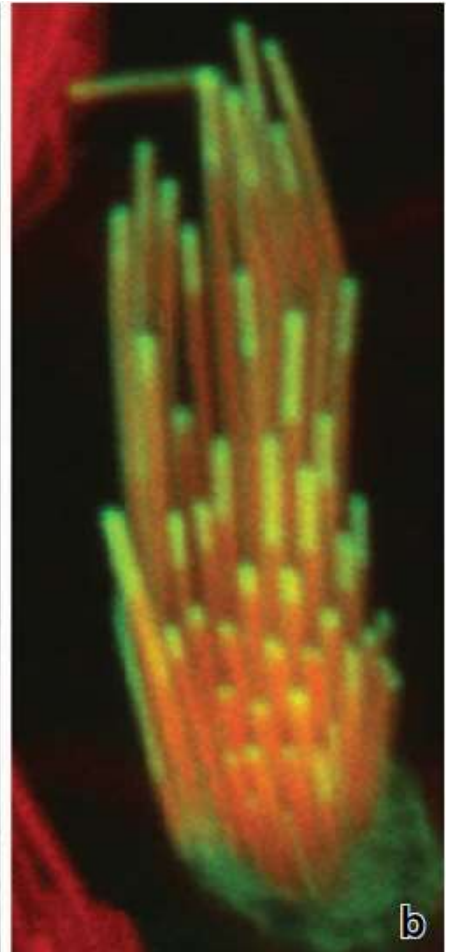
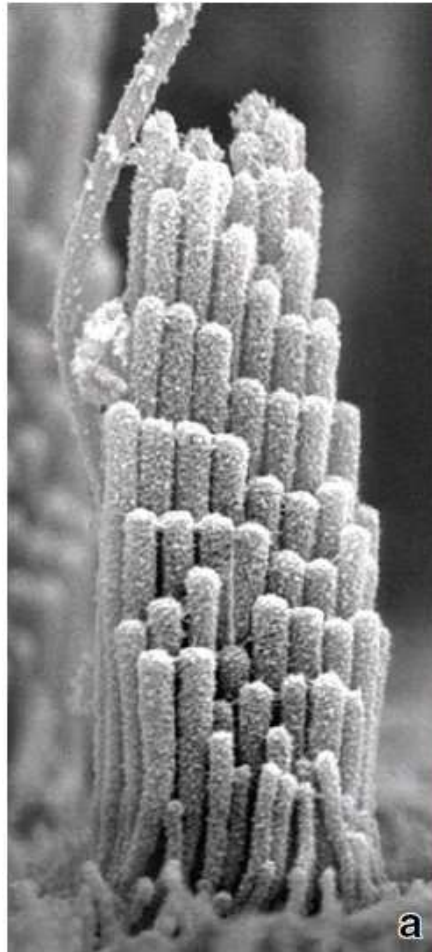
Microvilli and Stereocilia

	general structure	cross-section	motion trajectory	localization and function
microvilli	 Average 1-3 μm in length, bundle of actin filament anchored in the terminal web	 Core of actin filaments cross-linked by actin-bundling proteins; diameter 50-100 nm	 Passive movement due to contraction of terminal web	• present in many epithelial cells • increase absorptive surface of the cell • visible in LM as striated border (interstitial absorptive cells) or brush border (kidney tubule cells)
stereocilia	 epididymis inner ear Considerable longer; up to 120 μm, actin filament bundle anchored in the terminal web; capable of regeneration (inner ear)	 Core of actin filaments cross-linked by actin-bundling proteins; diameter 100-150 nm	 Passive movement due to fluid flow (genital system) or vibration of endolymph (inner ear)	• limited distribution • in male reproductive system (epididymis, proximal part of ductus deferens) have absorptive function • in sensory hair cells in the inner ear function as mechanoreceptors

Dynamic turnover of an internal architecture of stereocilia:

a. This scanning electron micrograph shows stereocilia of sensory epithelium of the inner ear. They are uniform in diameter and organized into ridged bundles of increasing heights. 47,000x.

b. Confocal microscopy image shows incorporation of the -actin green fluorescent protein (GFP) and espin-GFP to the tip of the stereocilia (green). Actin filaments in the core of the stereocilia are counterstained with Rhodamine/phalloidin (red).



TIP 1: Distinguishing Features of Epithelial Tissues

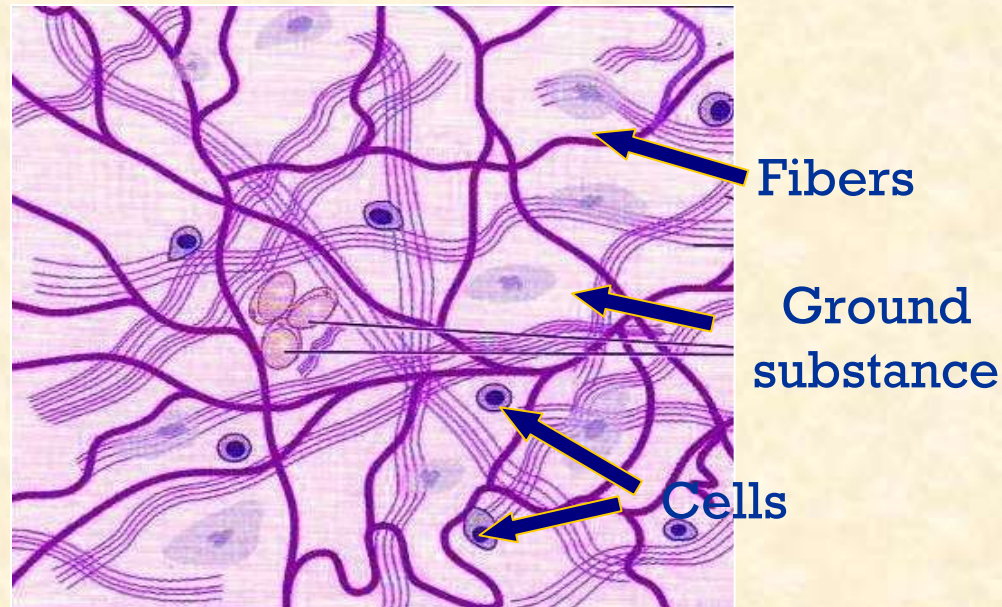
- 1. High cellularity**
- 2. Matrix paucity**
- 3. Avascularity**
- 4. Cohesiveness**
- 5. Cellular polarity**
- 6. Morphological diversity**
- 7. Superficial location**
- 8. Diverse physiological functions**
- 9. High Regenerative capability**
- 10. Constituting Parenchyma of organs and glands**
- 11. Good innervation**
- 12. Diverse embryological origin**

- **CONNECTIVE TISSUE**

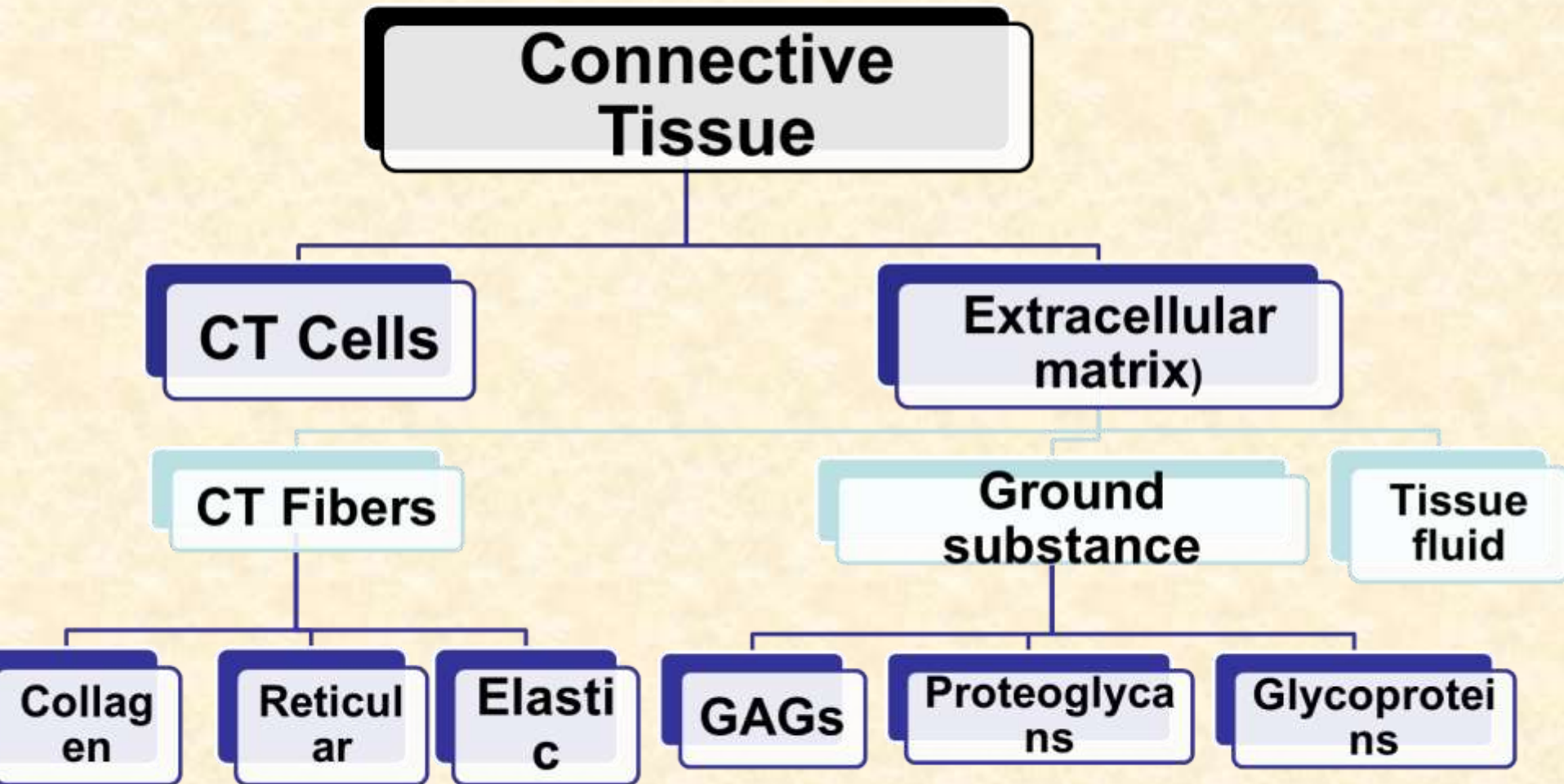
Connective tissue consists of **cells** widely separated from each other, and embedded in **extracellular matrix** consists of amorphous substance (**ground substance**) and **fibers**

Connective tissue **supports, binds and connects** various tissues and organs

Connective tissue is **rich in blood vessels**, lymphatics and nerves. It carries O₂ and nutrients to cells in various tissues of the body



ORGANIZATION of CT



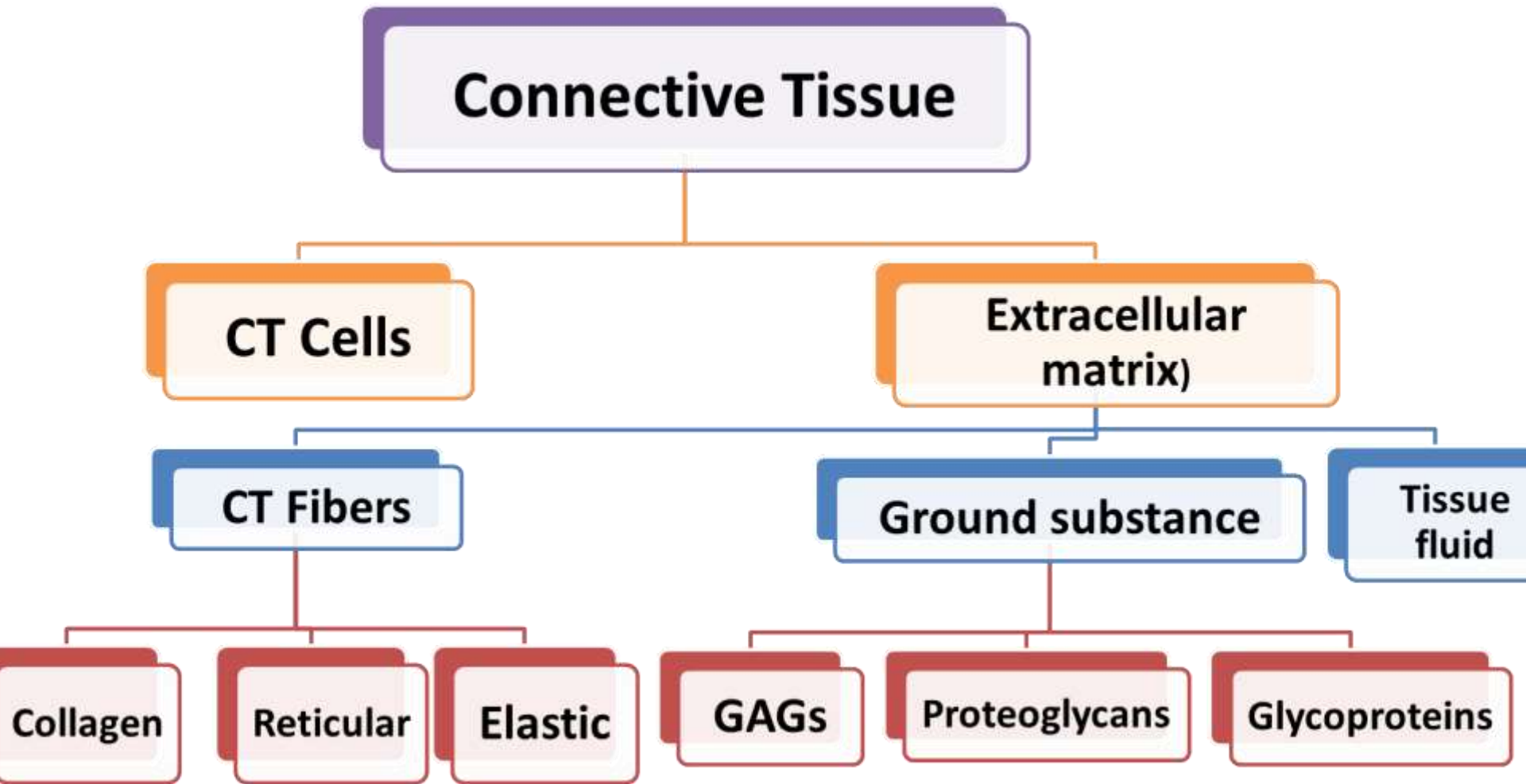
Distinguishing Features of Connective Tissues

1. Less cellularity
2. Matrix abundance
3. High vascularity
4. Morphological diversity
5. Wide distribution
6. Subepithelial location
7. Functional diversity: including structural, nutritive and metabolic supportive functions
8. Regenerative capability
9. Cellular variety
10. Constituting stroma of organs and glands
11. Heavy innervation
12. Mesenchymal origin

Epithelium and CT



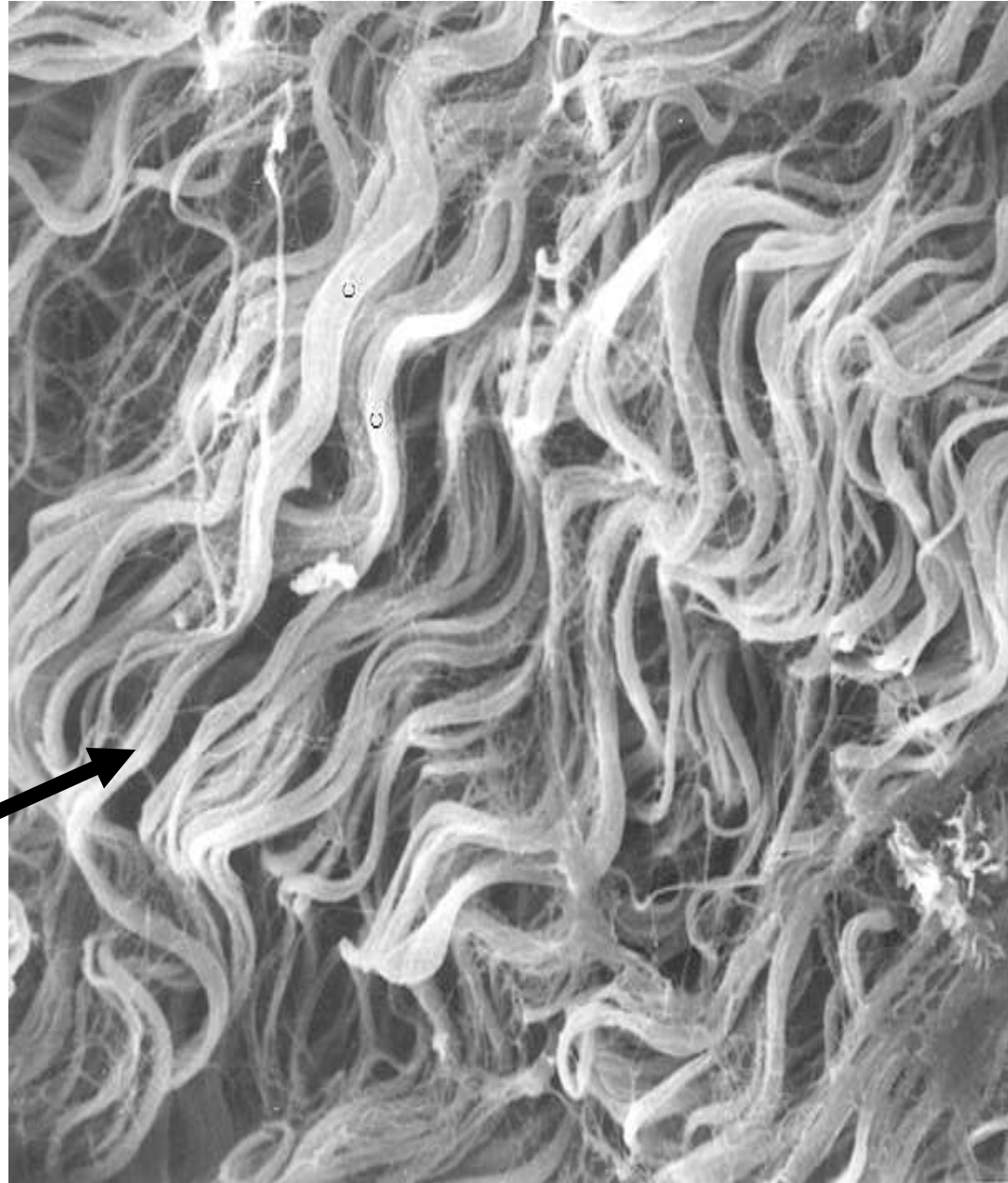
ORGANIZATION of CT



Fibers in Fibrous Connective Tissue:

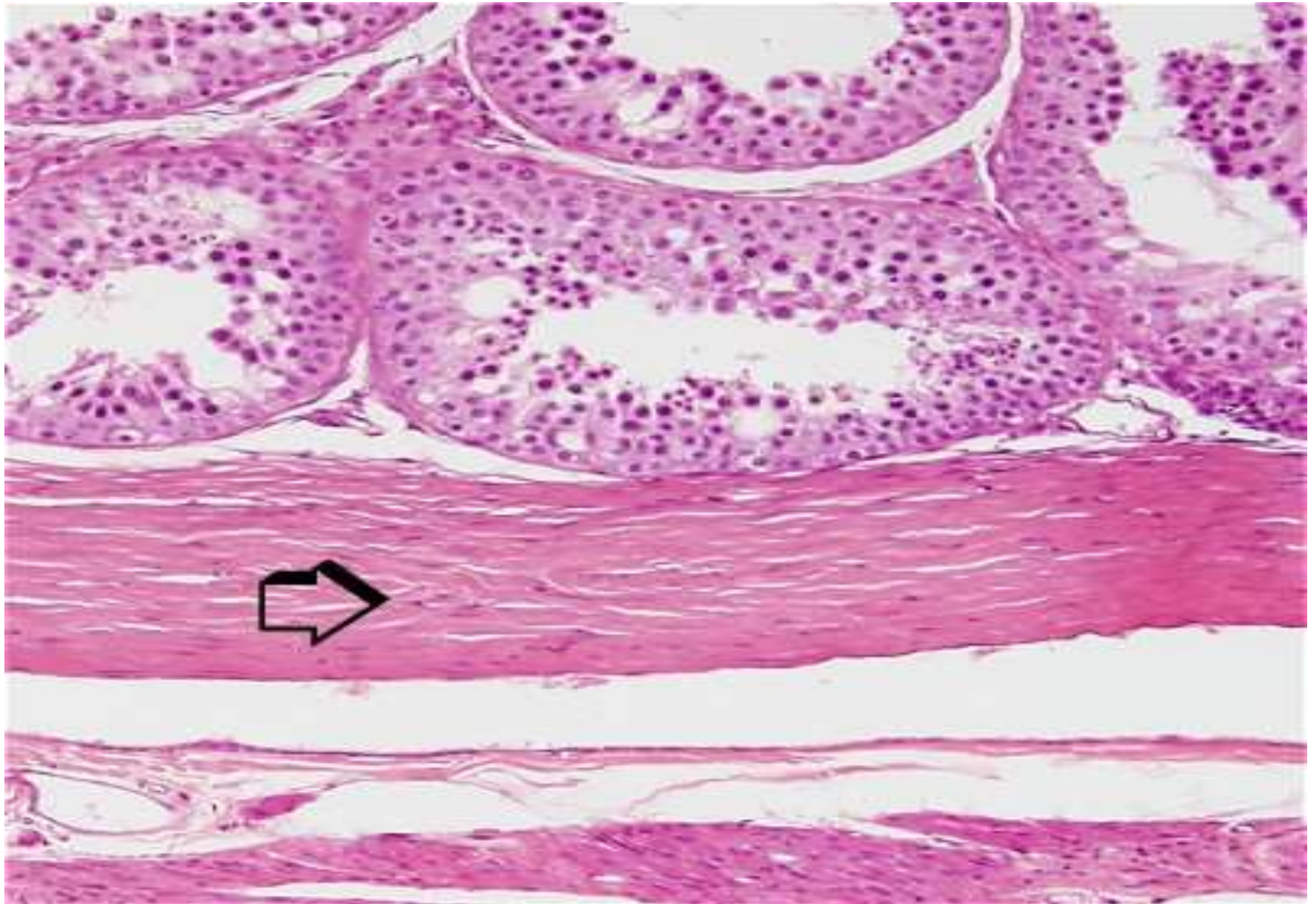
1. Collagenous Fibers
2. Reticular Fibers
3. Elastic Fibers

Collagen Fibers as
seen by scanning
electron microscope



1. Collagenous Fibers (KOL-a-jen; colla = glue, gen= produced)

- tough, firm, inelastic fibers which resist the tensile forces upon the tissues (flexible but with extremely high tensile strength), also called **white fibers**.
- the most common fibers in connective tissue and the most abundant proteins in the body.
- Although fresh collagen fibers are colorless strands, when present in large numbers (eg, in tendons) they appear white.
- The highly regular orientation of subunits in collagen fibers makes them birefringent under the polarizing microscope.
- In the light microscope, collagen fibers are acidophilic; they stain **pink** with eosin, **blue** with Mallory trichrome stain, **green** with Masson trichrome stain, and **red** with Sirius red.



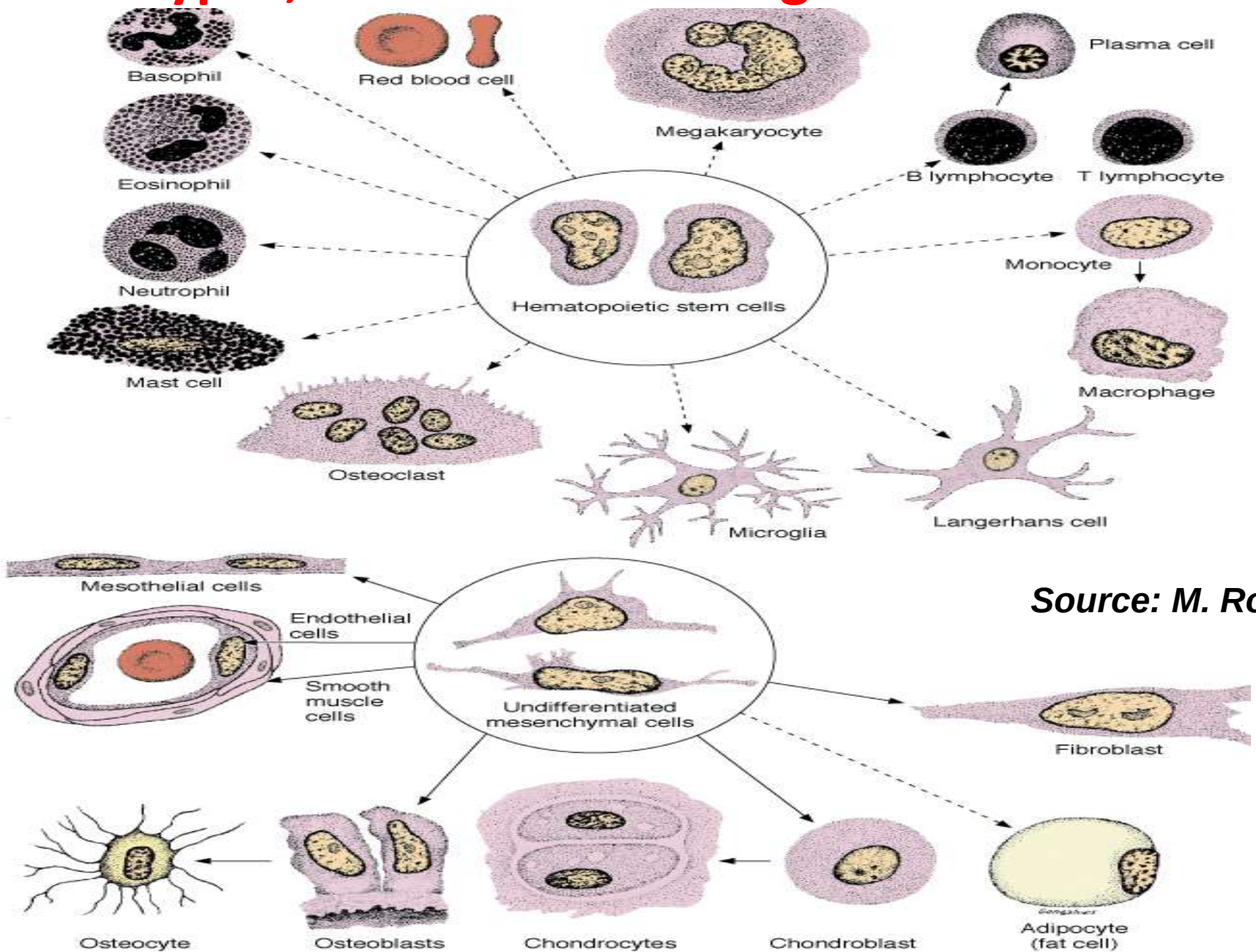
Blue stained collagenous tissue in Lamina propria



Collagenous Fibers

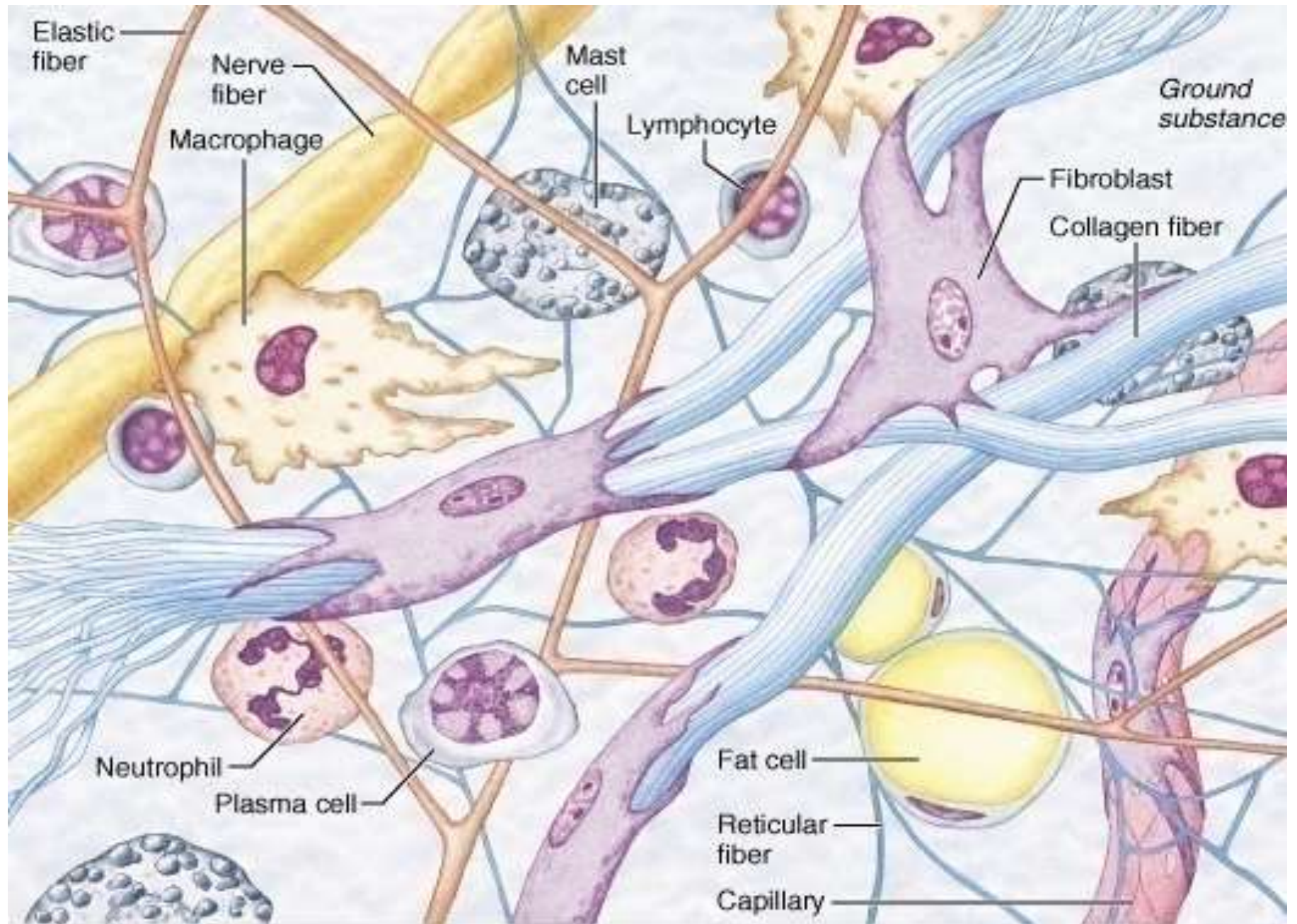
- synthesized by different types of cells in d/t tissues: **fibroblasts, chondroblasts, osteoblasts, odontoblasts, reticular cells**, adipocytes, **smooth muscle fibers**, etc
- ultrastructurally consists of:
 - Collagen macromolecules called **Tropocollagens** running parallel and overlapping each other.
 - **Tropocollagens form collagen fibrils**, each of which represents a structural unit (the **unit fibrils of collagen**).
 - Each unit fibril shows **transverse bands**.
 - Each tropocollagen molecule is **a right-handed triple helix** formed by intertwining of **three procollagen α -polypeptide chains**.

Types, Sources and Lineages of CT Cells

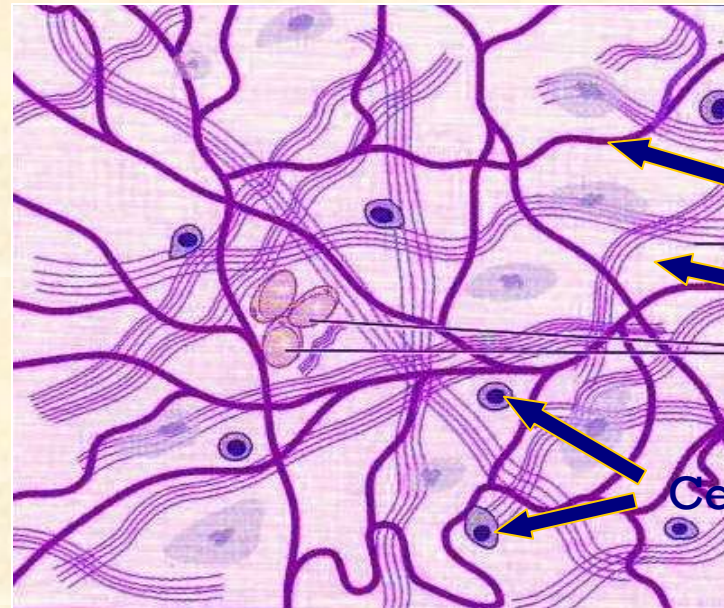
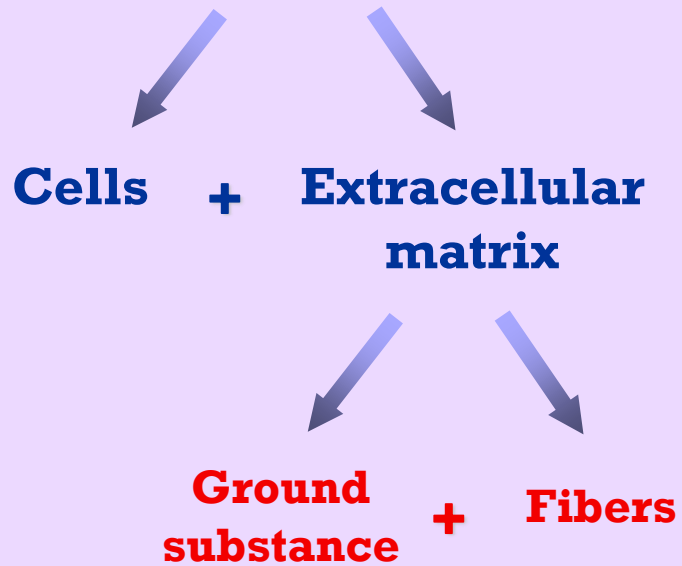


Source: M. Ross

Composition of ordinary connective tissue (CT Proper)



**Connective Tissue
consists of**



Fibers

**Ground
substance**

Cells